

// BYE-BYE LOCK-IN: AUTOMATE CLOUD SWITCHING USING OPEN SOURCE

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But Why?

Tool Stacks



PM+Docs



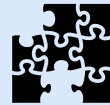
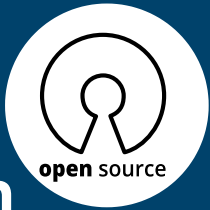
Software Dev



DevOps

...

EcoSystem



Integrations



AI Assistant



Low Ops

Infra



K8s

But Why?

Tool Stacks



PM+Docs

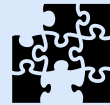
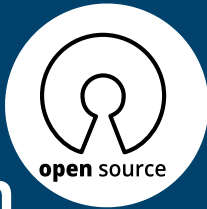


Software Dev

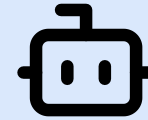


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VM



K8s

Automatic Migration



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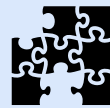
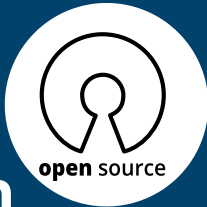
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K8s



Cloud
Providers



On
Prem



...

Automatic Migration



Automatic Cloud Switching

Reasons to switch

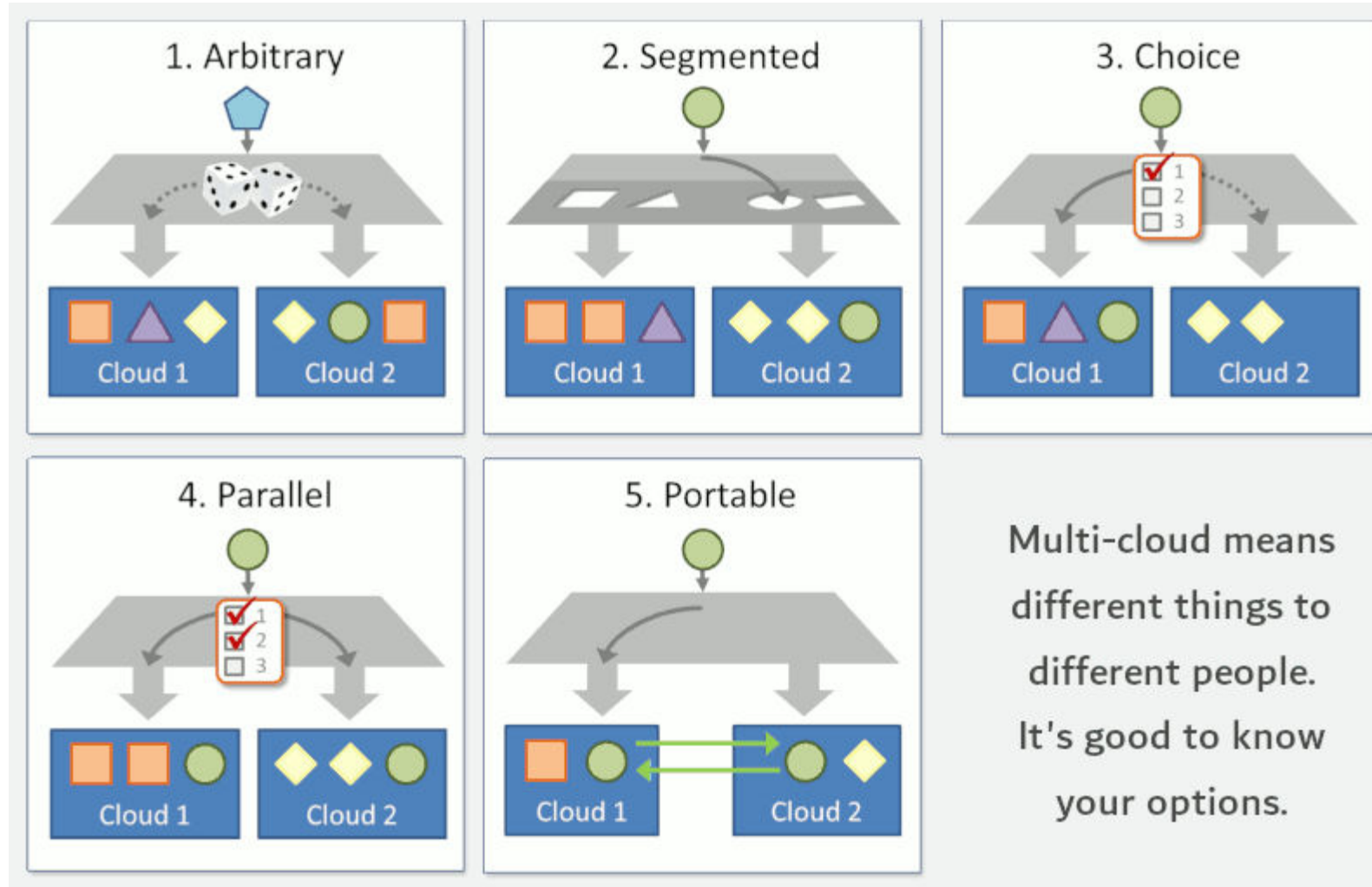
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- 💶 💶 💶 ➔ 💶
- ☁ ↔ 🏠
- ...

!! Ability to switch (digital sovereignty)

➡ Multi Cloud



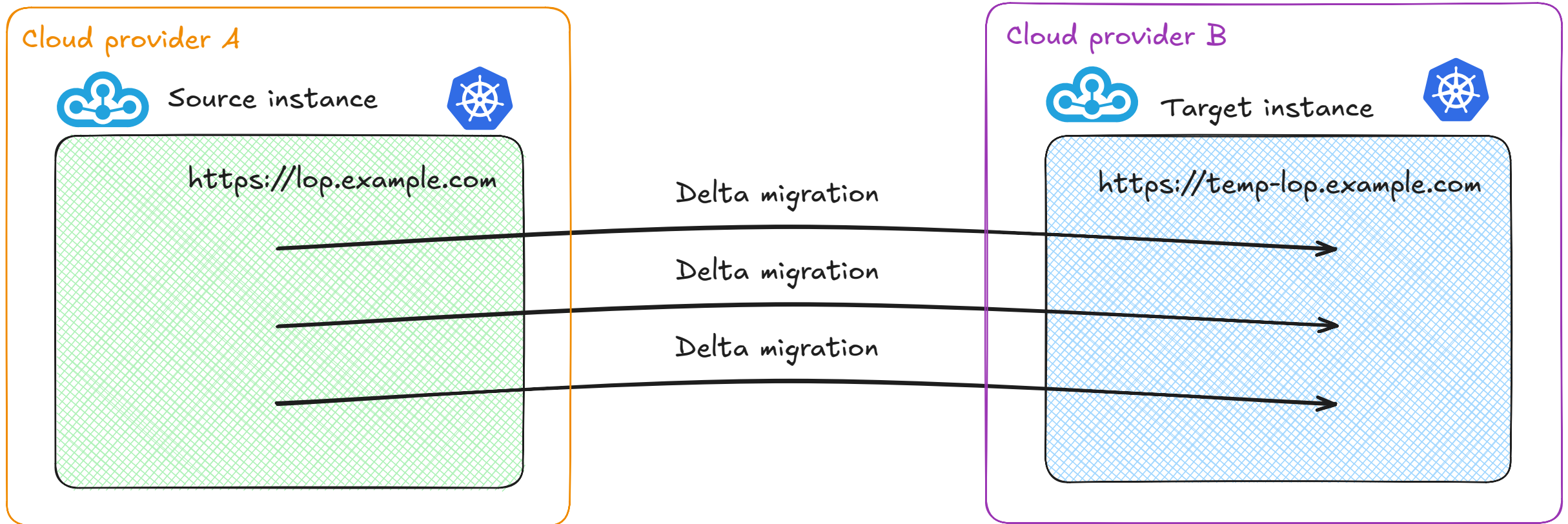
Multi Cloud Architecture Options



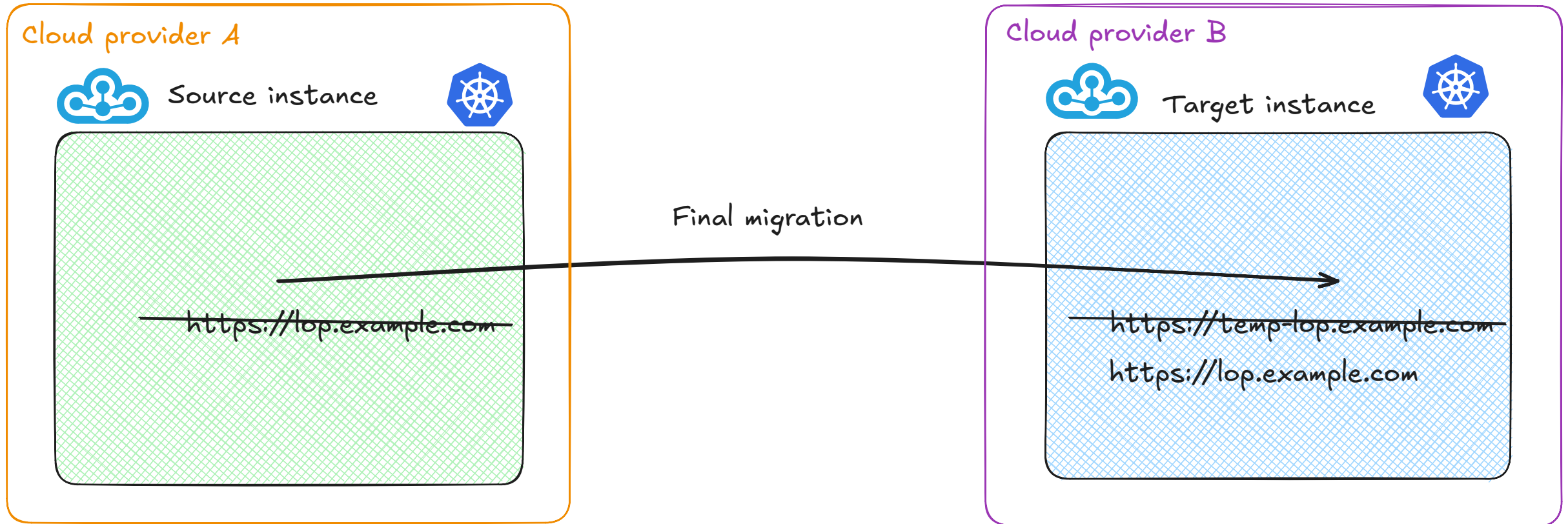
Automatic Cloud Switching



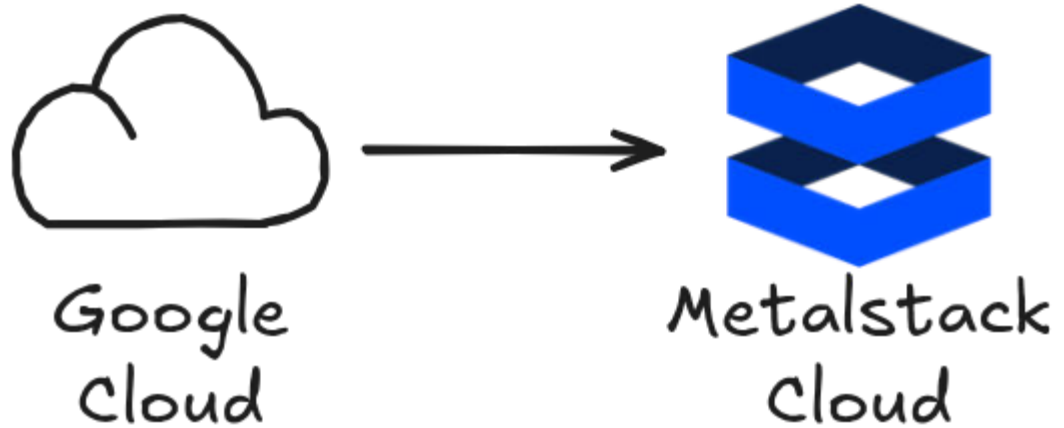
Delta migration



Final migration



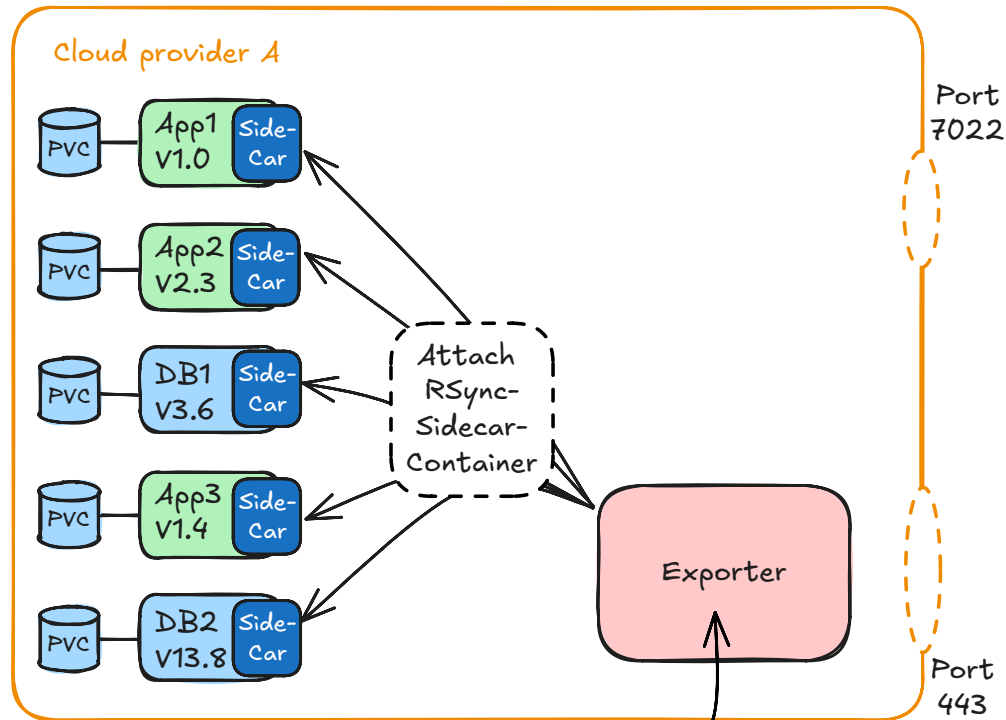
Demo migration

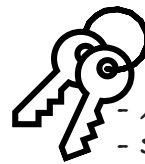


Preparing the migration



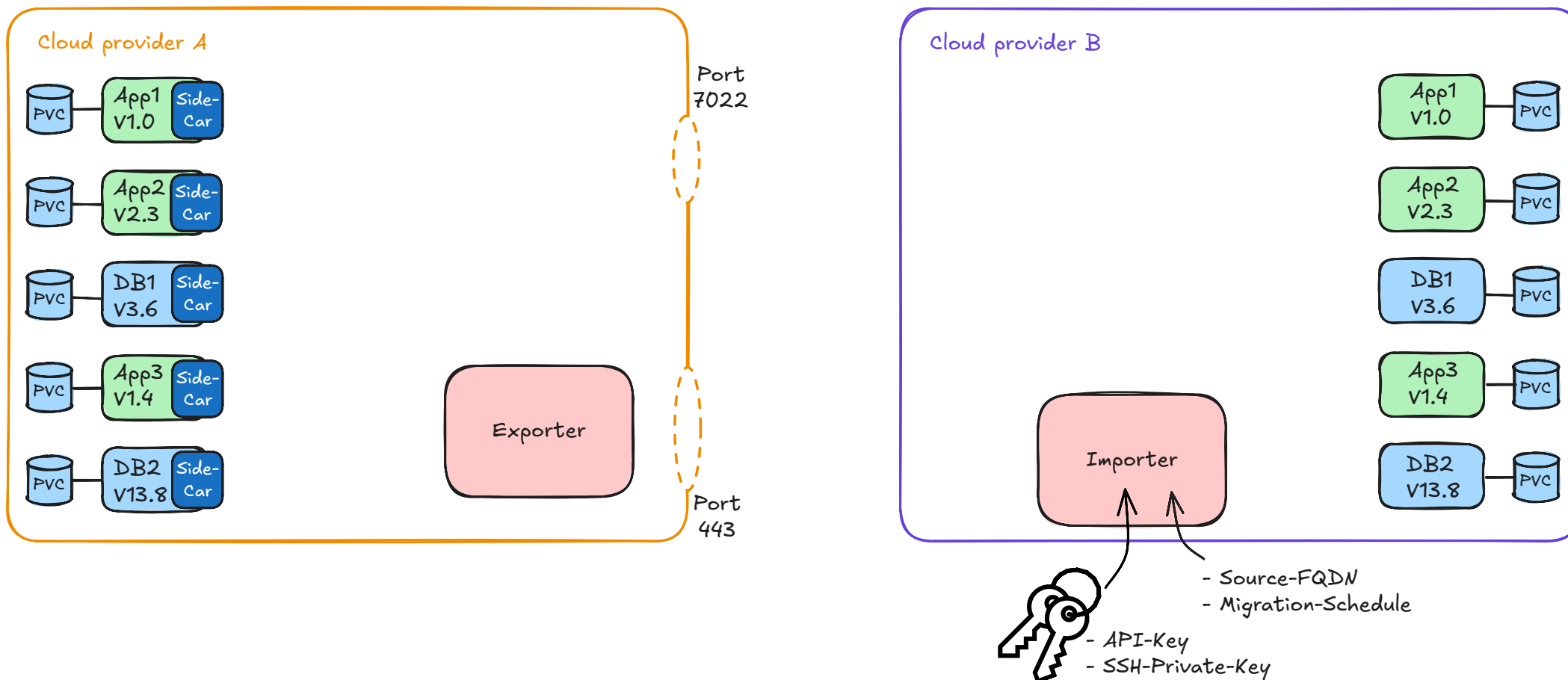
Prepare source instance



 - API-Key
- SSH-Public-Key



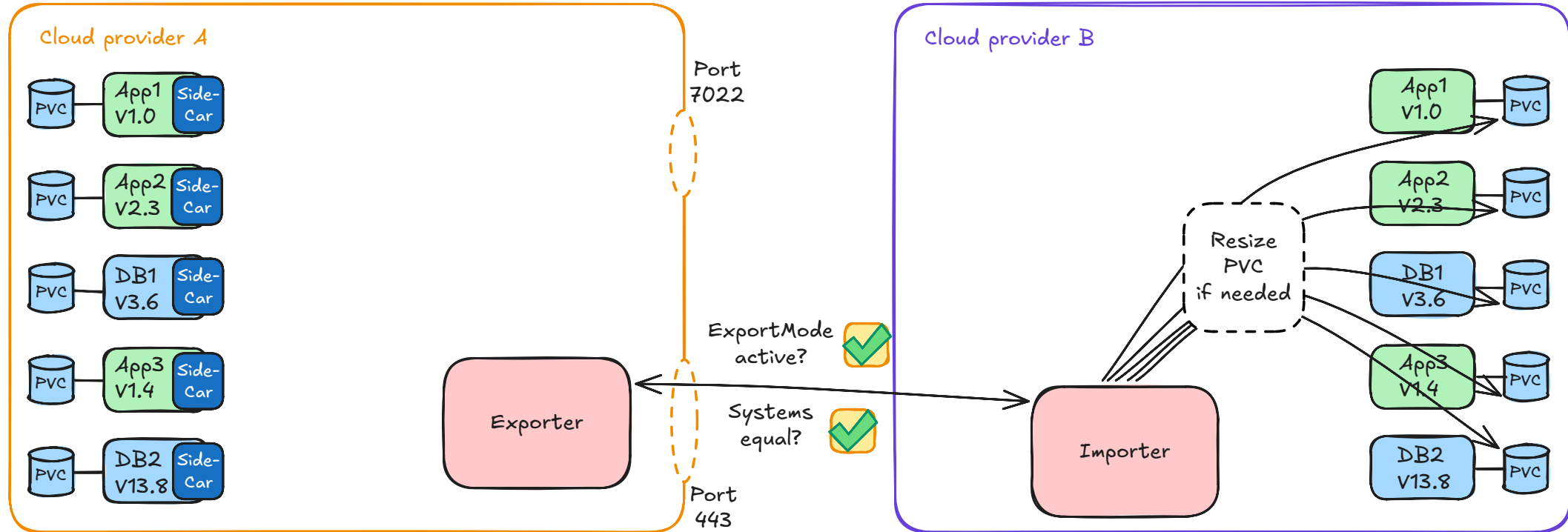
Prepare target instance



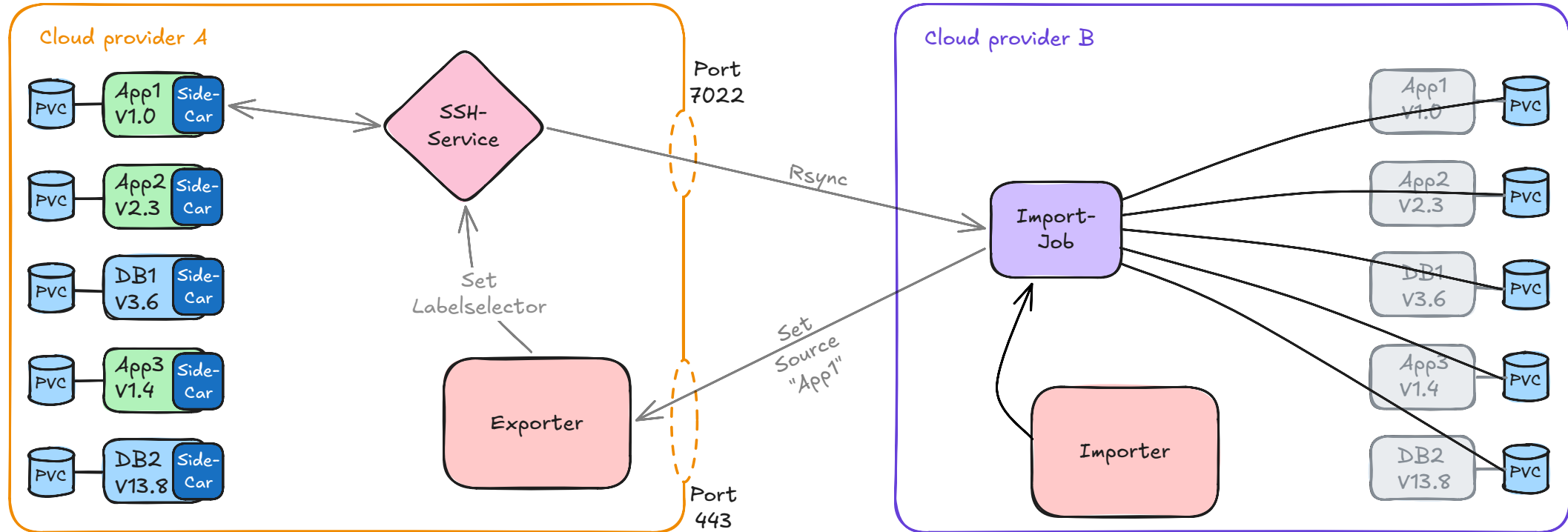
Executing the migration



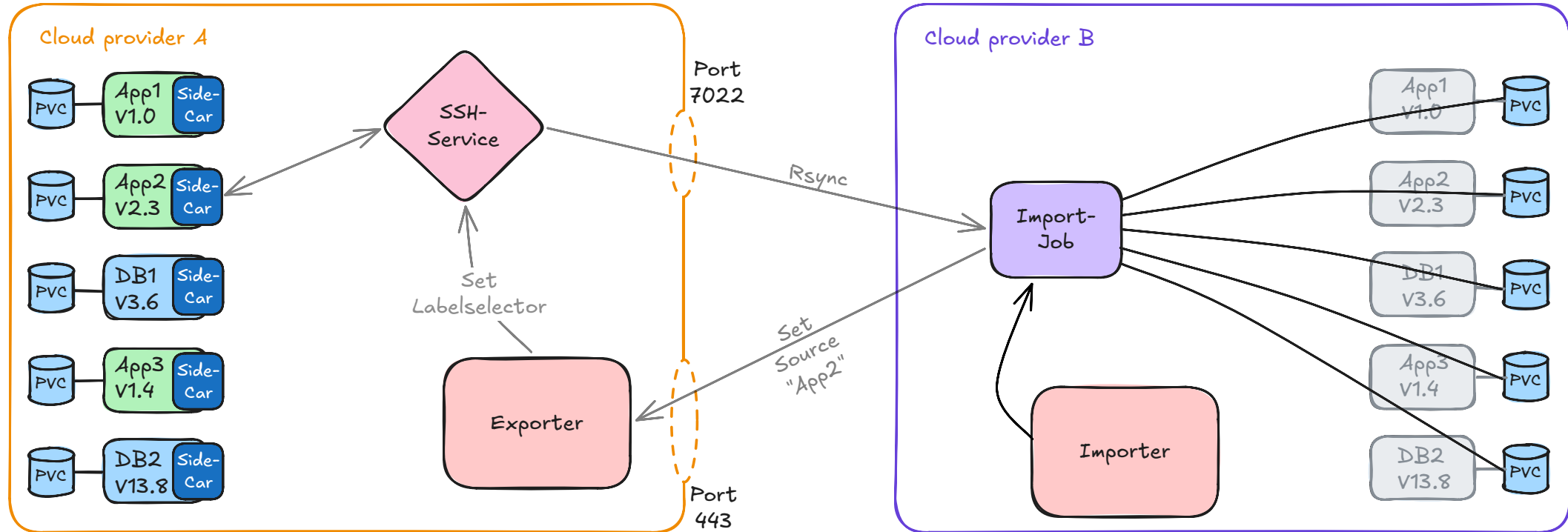
Verify preconditions



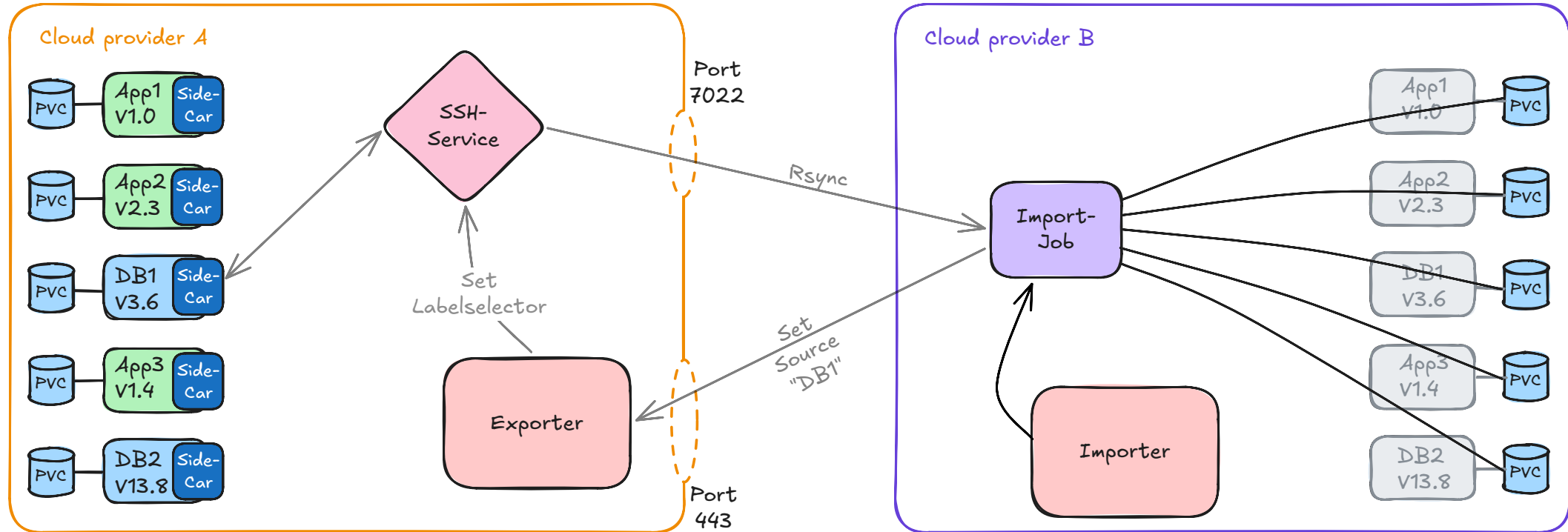
Migrate volume data



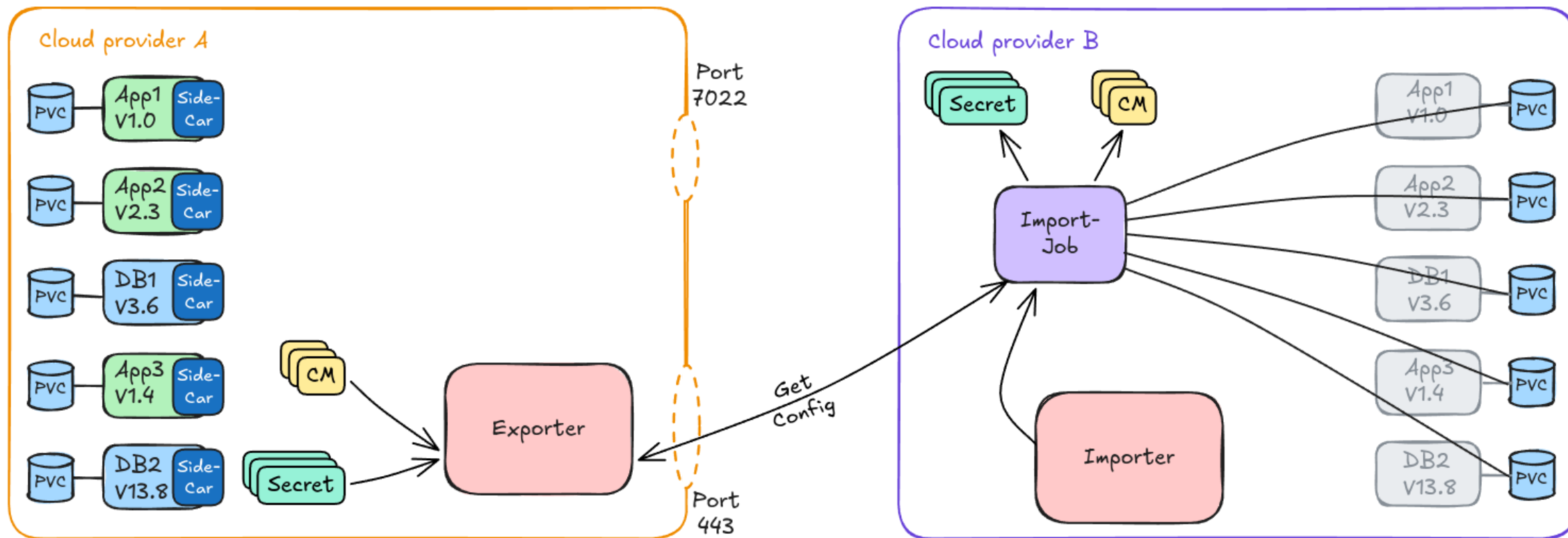
Migrate volume data



Migrate volume data



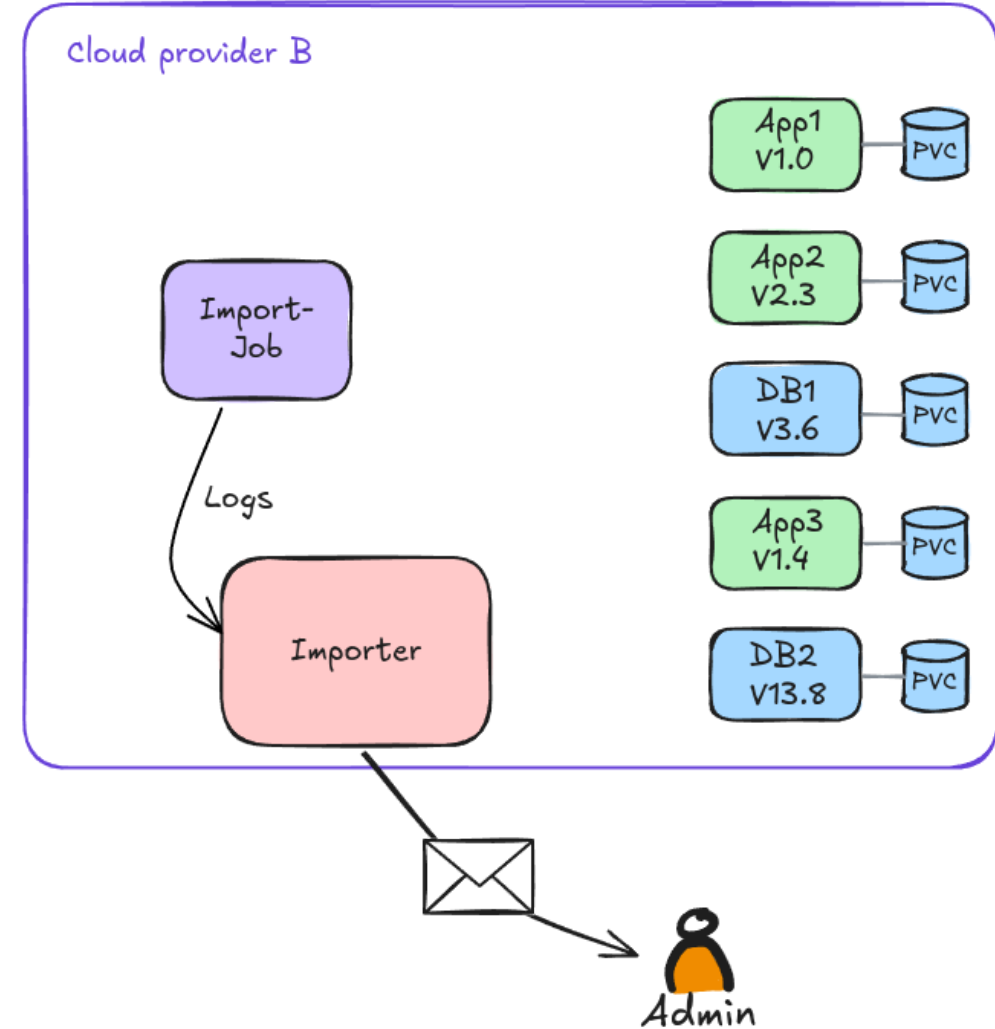
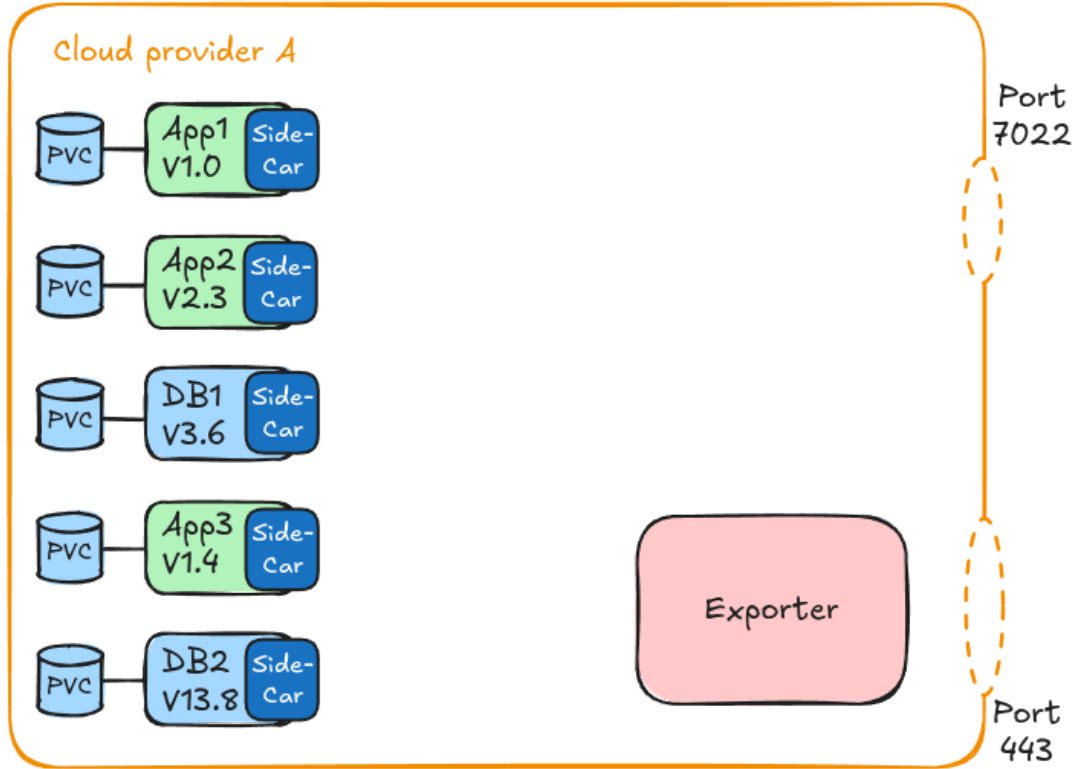
Migrate configuration data



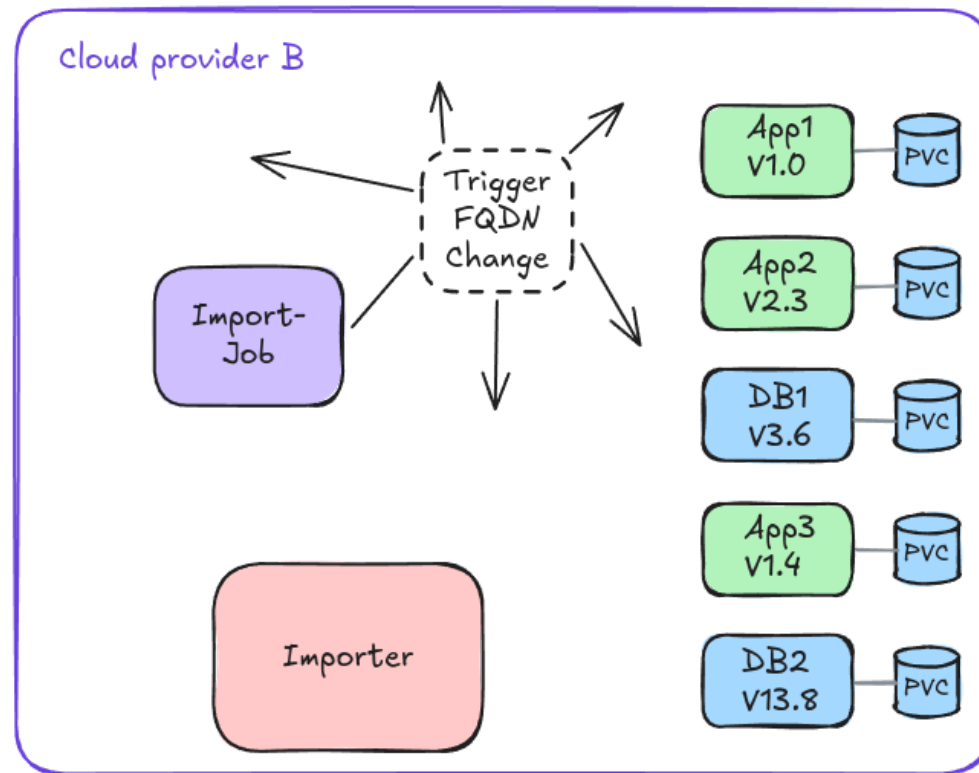
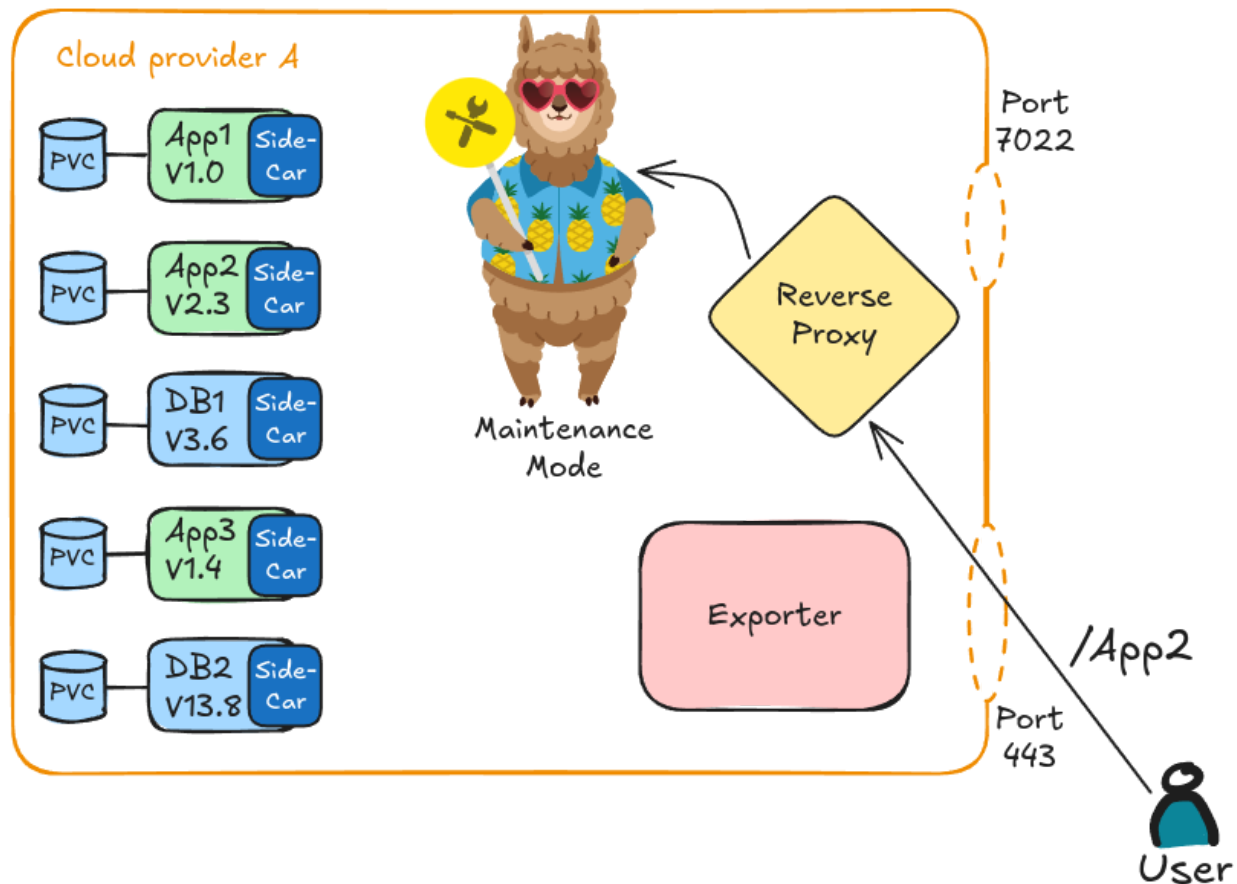
Finishing the migration



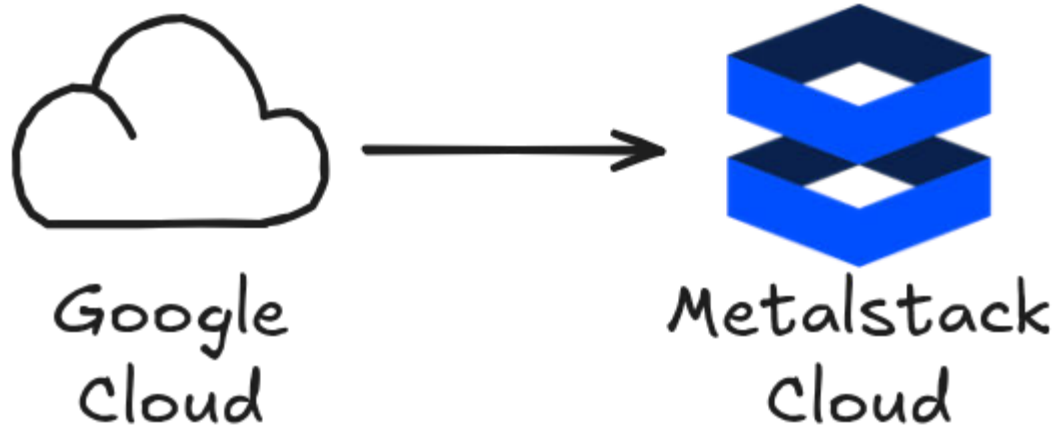
Finish migration



Additional steps on final migration



Demo migration



Conclusion



Migration without the usual pain

-  Repeatable migrations → safe to test
-  Delta-based transfer with rsync
-  Minimal network requirements (443 + SSH)

What we learned

-  PoCs helped de-risk key decisions
-  Separation (exporter / importer) kept it flexible
-  Manual testing became a bottleneck

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 cloudogu/ces-exporter  cloudogu/ces-importer

Image sources

- Automatic Cloud Switching
https://www.freepik.com/free-vector/cloud-network-technology-icon-white-gradient-background_16406372.htm
- Buzzwords
<https://makeameme.org/meme/buzzwords-buzzwords-everywhere-a120508e67>